TRIPPLE 7 MOVIE EMPIRE REVIEW

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1. OVERVIEW

Tripple 7 movie empire is a start up company that wants to compete in the movie industry. This analysis will help to give a general overview of the industry, to ensure the stakeholders understand how the industry works and what to expect of it, which will be ideal for decison making.

1.1 Business Problem

Tripple Seven plans to launch a new movie studio but lacks experience in the film industry. The company does not know which types of movies perform best at the box office or what factors contribute to a film's financial success.

Without data-driven insights, investing in movie production could lead to poor financial outcomes. To reduce this risk, the company needs to understand how factors such as movie genre, release timing, and audience engagement affect revenue and profitability.

This project aims to analyze historical movie data to identify successful patterns and provide clear recommendations that will help Tripple Seven make informed decisions on the types of films to produce.

1.2 Business Understanding

Tripple 7 movie empire is a multi-billion dollar industry that has been growing rapidly in recent years due to various factors that have ensured a steady rise from the analog to digital eras.

We've witnessed improvements in distribution, quality, genres, acting, among other things.

One of the major factors has been the internet, which today is easily accessible to majority of the population worldwide. This has lead to the introduction of various streaming platforms, making it easy for viewers to download any type of movies they want to watch. Hence we have migrated from eras where we had to use compact disks(CD's/DVD's) which needed a physical store to buy them, to now using streaming platforms at the comfort of your home without needing to go out and get access to a particular movie, as long as you have internet access.

Other factors have also majorly contributed to its growth, ie,

- The use of various visual effects like CGI,VFX ensuring the movie is of high quality.
- Large investments to ensure the logistics in a movie are met and the movie is finished and distributed on time.
- Changing audience preferences ie genre preferences and cultural diversity has lead to content that caters to various audience.
- Consumer spending due to strong economy can lead to high box office revenues.
- Production and creative factors ie, script quality, acting skills, marketing and promotions can generate great revenues.

However, all industries have their own risks that can impact various aspects in production like piracy, economic downturns, budget overruns, changing audience preferences, weather-related risks, casts illness/injury, poor box-office run, among other things.Therefore it is necessary to try and mitigate all/most of these risks during the planning and production periods. One of the main ways to reduce risks is by factoring data analysis to know the current trends and target audience.

1.3 Objective

1. To determine the movie genres that are the most profitable.
2. To determine the best time to release a movie for maximum returns.
3. To determine whether higher-rated movies generate more revenue.

1.4 Business Questions

1. Which movie genres are the most profitable?
2. When is the best time to release a movie for maximum returns?
3. Do higher-rated movies generate more revenue?

1.5 Dataset Dictonary

This dataset contains the following variables: ● **Unnamed**: 0: Index column created during data export (not meaningful for analysis).

● **genre_ids**: List of genre IDs associated with the movie.

● **id_x**: Movie ID from the first dataset (e.g., TMDB dataset).

● **original_language**: Original language in which the movie was produced (e.g., en).

● **original_title**: Original title of the movie.

● **popularity**: Popularity score based on user engagement and activity.

● **release_date_x**: Movie release date from the first dataset (format: YYYY-MM-DD).

● **title**: Movie title (standardized version).

● **vote_average**: Average user rating (scale of 0–10).

● **vote_count**: Total number of user votes received.

● **id_y**: Movie ID from the second dataset after merging.

● **release_date_y**: Movie release date from the second dataset (text format, e.g., Mar 26, 2010).

● **movie**: Movie title from the second dataset.

● **production_budget**: Total budget used to produce the movie (in USD, stored as a string with $).

● **domestic_gross**: Revenue generated within the domestic market (USD).

● **worldwide_gross**: Total global revenue generated by the movie (USD).

2. Data Loading and Exploration

2.1 Loading dataset with necessary libraries:

In [1]:
```
#Importing the necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

2.2 chosen Dataset

In [2]:
```
#relevant dataset selected
empire_1 = pd.read_csv('tmdb.movies.csv')
empire_2 = pd.read_csv('tn.movie_budgets.csv')
#give reason why we choose the 2
```

The columns in these two datasets compliments each other and are relevant for our analysis as well as answers our business questions.

2.3 Merging two chosen Dataset

In [3]:
```
#merging the two dataset
merged_empire = pd.merge(
    empire_1,
    empire_2,
    left_on='original_title',
    right_on='movie',
    how='inner')
merged_empire
```

Out[3]:

	Unnamed: 0	genre_ids	id_x	original_language	original_title	popularity	release_date_x	title	vote_average	vote_count	id_y	release_date_y	movie	production_budget	domestic_gross	worldwide_gross
0	1	[14, 12, 16, 10751]	10191	en	How to Train Your Dragon	28.734	2010-03-26	How to Train Your Dragon	7.7	7610	30	Mar 26, 2010	How to Train Your Dragon	$165,000,000	$217,581,232	$494,870,992
1	2	[12, 28, 878]	10138	en	Iron Man 2	28.515	2010-05-07	Iron Man 2	6.8	12368	15	May 7, 2010	Iron Man 2	$170,000,000	$312,433,331	$621,156,389
2	3	[16, 35, 10751]	862	en	Toy Story	28.005	1995-11-22	Toy Story	7.9	10174	37	Nov 22, 1995	Toy Story	$30,000,000	$191,796,233	$364,545,516
3	2473	[16, 35, 10751]	862	en	Toy Story	28.005	1995-11-22	Toy Story	7.9	10174	37	Nov 22, 1995	Toy Story	$30,000,000	$191,796,233	$364,545,516
4	4	[28, 878, 12]	27205	en	Inception	27.920	2010-07-16	Inception	8.3	22186	38	Jul 16, 2010	Inception	$160,000,000	$292,576,195	$835,524,642
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2311	26323	[]	509316	en	The Box	0.600	2018-03-04	The Box	8.0	1	66	Nov 6, 2009	The Box	$25,000,000	$15,051,977	$34,356,760
2312	26425	[10402]	509306	en	The Box	0.600	2018-03-04	The Box	6.0	1	66	Nov 6, 2009	The Box	$25,000,000	$15,051,977	$34,356,760
2313	26092	[35, 16]	546674	en	Enough	0.719	2018-03-22	Enough	8.7	3	68	May 24, 2002	Enough	$38,000,000	$39,177,215	$50,970,660
2314	26322	[]	513161	en	Undiscovered	0.600	2018-04-07	Undiscovered	8.0	1	7	Aug 26, 2005	Undiscovered	$9,000,000	$1,069,318	$1,069,318
2315	26508	[16]	514492	en	Jaws	0.600	2018-05-29	Jaws	0.0	1	41	Jun 20, 1975	Jaws	$12,000,000	$260,000,000	$470,700,000

2316 rows × 16 columns

We combine and merge the release dates(x,y) in both datasets to just release_date and dropping the other columns

In [4]:
```
merged_empire['release_date'] = merged_empire['release_date_x'].combine_first(merged_empire['release_date_y'])
```

In [5]:
```
merged_empire = merged_empire.drop(columns=['release_date_x', 'release_date_y'])
```

2.4 Previewing the data

In [6]:
```
merged_empire.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 2316 entries, 0 to 2315
Data columns (total 15 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Unnamed: 0         2316 non-null   int64
 1   genre_ids          2316 non-null   object
 2   id_x               2316 non-null   int64
 3   original_language  2316 non-null   object
 4   original_title     2316 non-null   object
 5   popularity         2316 non-null   float64
 6   title              2316 non-null   object
 7   vote_average       2316 non-null   float64
 8   vote_count         2316 non-null   int64
 9   id_y               2316 non-null   int64
 10  movie              2316 non-null   object
 11  production_budget  2316 non-null   object
 12  domestic_gross     2316 non-null   object
 13  worldwide_gross    2316 non-null   object
 14  release_date       2316 non-null   object
dtypes: float64(2), int64(4), object(9)
memory usage: 289.5+ KB
```

Converting the revenue columns from object type to numerical type

In [7]:
```
#Converting multiple columns at once
revenue_col = ['production_budget', 'domestic_gross', 'worldwide_gross']

for revenue in revenue_col:
   merged_empire[revenue] = merged_empire[revenue].str.replace('$', '', regex=False)\
                    .str.replace(',', '', regex=False)\
                    .astype(float)
```

In [8]:
```
merged_empire.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 2316 entries, 0 to 2315
Data columns (total 15 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Unnamed: 0         2316 non-null   int64
 1   genre_ids          2316 non-null   object
 2   id_x               2316 non-null   int64
 3   original_language  2316 non-null   object
 4   original_title     2316 non-null   object
 5   popularity         2316 non-null   float64
 6   title              2316 non-null   object
 7   vote_average       2316 non-null   float64
 8   vote_count         2316 non-null   int64
 9   id_y               2316 non-null   int64
 10  movie              2316 non-null   object
 11  production_budget  2316 non-null   float64
 12  domestic_gross     2316 non-null   float64
 13  worldwide_gross    2316 non-null   float64
 14  release_date       2316 non-null   object
```

```
dtypes: float64(5), int64(4), object(6)
memory usage: 289.5+ KB
```

EDA - Data Cleaning

In [9]:
```
#checking null values
merged_empire.isna().sum()
```

Out[9]:
```
Unnamed: 0           0
genre_ids            0
id_x                 0
original_language    0
original_title       0
popularity           0
title                0
vote_average         0
vote_count           0
id_y                 0
movie                0
production_budget    0
domestic_gross       0
worldwide_gross      0
release_date         0
dtype: int64
```

The merged data is complete we have no missing values

In [10]:
```
#checking for duplicates
merged_empire.duplicated().value_counts()
```

Out[10]:
```
False    2316
dtype: int64
```

The data also has no duplicated values

- Creating a dictionary by combining the key(genre_id) and value(genre)

In [11]:
```
# TMDb genre mapping
genre_mapping = {
    28: "Action",
    12: "Adventure",
    16: "Animation",
    35: "Comedy",
    80: "Crime",
    99: "Documentary",
    18: "Drama",
    10751: "Family",
    14: "Fantasy",
    36: "History",
    27: "Horror",
    10402: "Music",
    9648: "Mystery",
    10749: "Romance",
    878: "Science Fiction",
    10770: "TV Movie",
    53: "Thriller",
    10752: "War",
    37: "Western"
}

# Convert genre_ids from string to list if needed
merged_empire['genre_ids'] = merged_empire['genre_ids'].apply(lambda x: eval(x) if isinstance(x, str) else x)

# Map all IDs to names
merged_empire['all_genres'] = merged_empire['genre_ids'].apply(lambda ids: [genre_mapping.get(i) for i in ids])

# Optional: Convert list to comma-separated string for readability
merged_empire['all_genres_str'] = merged_empire['all_genres'].apply(lambda x: ', '.join([g for g in x if g]))

merged_empire[['original_title', 'genre_ids', 'all_genres_str']].head(10)
```

Out[11]:

	original_title	genre_ids	all_genres_str
0	How to Train Your Dragon	[14, 12, 16, 10751]	Fantasy, Adventure, Animation, Family
1	Iron Man 2	[12, 28, 878]	Adventure, Action, Science Fiction
2	Toy Story	[16, 35, 10751]	Animation, Comedy, Family
3	Toy Story	[16, 35, 10751]	Animation, Comedy, Family
4	Inception	[28, 878, 12]	Action, Science Fiction, Adventure
5	Percy Jackson & the Olympians: The Lightning T...	[12, 14, 10751]	Adventure, Fantasy, Family
6	Avatar	[28, 12, 14, 878]	Action, Adventure, Fantasy, Science Fiction
7	Toy Story 3	[16, 10751, 35]	Animation, Family, Comedy
8	Despicable Me	[16, 10751, 35]	Animation, Family, Comedy
9	Megamind	[16, 28, 35, 10751, 878]	Animation, Action, Comedy, Family, Science Fic...

In [12]:
```
# Describing the dataset to get the summary statistics for the merged empire

merged_empire.describe()
```

Out[12]:

	Unnamed: 0	id_x	popularity	vote_average	vote_count	id_y	production_budget	domestic_gross	worldwide_gross
count	2316.000000	2316.000000	2316.000000	2316.000000	2316.000000	2316.000000	2.316000e+03	2.316000e+03	2.316000e+03
mean	11447.636874	210410.569516	10.685747	6.206261	1712.143351	50.842832	3.790108e+07	4.953869e+07	1.201667e+08
std	7557.377373	144548.546190	8.244311	1.178079	2704.578576	28.359866	5.150964e+07	8.261483e+07	2.218496e+08
min	1.000000	62.000000	0.600000	0.000000	1.000000	1.000000	9.000000e+03	0.000000e+00	0.000000e+00
25%	5208.500000	71862.750000	5.487000	5.600000	63.000000	27.000000	5.000000e+06	7.788272e+05	3.224914e+06
50%	11160.500000	208501.500000	9.506000	6.300000	615.000000	51.000000	1.900000e+07	2.015790e+07	3.722812e+07
75%	17541.250000	330641.750000	14.423750	6.900000	2100.500000	75.000000	4.650000e+07	5.830223e+07	1.248731e+08
max	26508.000000	587627.000000	80.773000	10.000000	22186.000000	100.000000	4.250000e+08	7.605076e+08	2.776345e+09

In [13]:
```
merged_empire
```

Out[13]:

	Unnamed: 0	genre_ids	id_x	original_language	original_title	popularity	title	vote_average	vote_count	id_y	movie	production_budget	domestic_gross	worldwide_gross	release_date	all_genres	all_genres_str
0	1	[14, 12, 16, 10751]	10191	en	How to Train Your Dragon	28.734	How to Train Your Dragon	7.7	7610	30	How to Train Your Dragon	165000000.0	217581232.0	494870992.0	2010-03-26	[Fantasy, Adventure, Animation, Family]	Fantasy, Adventure, Animation, Family
1	2	[12, 28, 878]	10138	en	Iron Man 2	28.515	Iron Man 2	6.8	12368	15	Iron Man 2	170000000.0	312433331.0	621156389.0	2010-05-07	[Adventure, Action,	Adventure, Action,

	Unnamed: 0	genre_ids	id_x	original_language	original_title	popularity	title	vote_average	vote_count	id_y	movie	production_budget	domestic_gross	worldwide_gross	release_date	all_genres	all_genres_str
																Science Fiction]	Science Fiction
2	3	[16, 35, 10751]	862	en	Toy Story	28.005	Toy Story	7.9	10174	37	Toy Story	30000000.0	191796233.0	364545516.0	1995-11-22	[Animation, Comedy, Family]	Animation, Comedy, Family
3	2473	[16, 35, 10751]	862	en	Toy Story	28.005	Toy Story	7.9	10174	37	Toy Story	30000000.0	191796233.0	364545516.0	1995-11-22	[Animation, Comedy, Family]	Animation, Comedy, Family
4	4	[28, 878, 12]	27205	en	Inception	27.920	Inception	8.3	22186	38	Inception	160000000.0	292576195.0	835524642.0	2010-07-16	[Action, Science Fiction, Adventure]	Action, Science Fiction, Adventure
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2311	26323	[]	509316	en	The Box	0.600	The Box	8.0	1	66	The Box	25000000.0	15051977.0	34356760.0	2018-03-04	[]	
2312	26425	[10402]	509306	en	The Box	0.600	The Box	6.0	1	66	The Box	25000000.0	15051977.0	34356760.0	2018-03-04	[Music]	Music
2313	26092	[35, 16]	546674	en	Enough	0.719	Enough	8.7	3	68	Enough	38000000.0	39177215.0	50970660.0	2018-03-22	[Comedy, Animation]	Comedy, Animation
2314	26322	[]	513161	en	Undiscovered	0.600	Undiscovered	8.0	1	7	Undiscovered	9000000.0	1069318.0	1069318.0	2018-04-07	[]	
2315	26508	[16]	514492	en	Jaws	0.600	Jaws	0.0	1	41	Jaws	12000000.0	260000000.0	470700000.0	2018-05-29	[Animation]	Animation

2316 rows × 17 columns

In [14]:
```
# Creating a new column called 'total_profit'

merged_empire = merged_empire.assign(total_profit = merged_empire['worldwide_gross'] - merged_empire['production_budget'])
merged_empire
```

Out[14]:

	Unnamed: 0	genre_ids	id_x	original_language	original_title	popularity	title	vote_average	vote_count	id_y	movie	production_budget	domestic_gross	worldwide_gross	release_date	all_genres	all_genres_str	t
0	1	[14, 12, 16, 10751]	10191	en	How to Train Your Dragon	28.734	How to Train Your Dragon	7.7	7610	30	How to Train Your Dragon	165000000.0	217581232.0	494870992.0	2010-03-26	[Fantasy, Adventure, Animation, Family]	Fantasy, Adventure, Animation, Family	3
1	2	[12, 28, 878]	10138	en	Iron Man 2	28.515	Iron Man 2	6.8	12368	15	Iron Man 2	170000000.0	312433331.0	621156389.0	2010-05-07	[Adventure, Action, Science Fiction]	Adventure, Action, Science Fiction	4
2	3	[16, 35, 10751]	862	en	Toy Story	28.005	Toy Story	7.9	10174	37	Toy Story	30000000.0	191796233.0	364545516.0	1995-11-22	[Animation, Comedy, Family]	Animation, Comedy, Family	3
3	2473	[16, 35, 10751]	862	en	Toy Story	28.005	Toy Story	7.9	10174	37	Toy Story	30000000.0	191796233.0	364545516.0	1995-11-22	[Animation, Comedy, Family]	Animation, Comedy, Family	3
4	4	[28, 878, 12]	27205	en	Inception	27.920	Inception	8.3	22186	38	Inception	160000000.0	292576195.0	835524642.0	2010-07-16	[Action, Science Fiction, Adventure]	Action, Science Fiction, Adventure	6
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
2311	26323	[]	509316	en	The Box	0.600	The Box	8.0	1	66	The Box	25000000.0	15051977.0	34356760.0	2018-03-04	[]		
2312	26425	[10402]	509306	en	The Box	0.600	The Box	6.0	1	66	The Box	25000000.0	15051977.0	34356760.0	2018-03-04	[Music]	Music	
2313	26092	[35, 16]	546674	en	Enough	0.719	Enough	8.7	3	68	Enough	38000000.0	39177215.0	50970660.0	2018-03-22	[Comedy, Animation]	Comedy, Animation	
2314	26322	[]	513161	en	Undiscovered	0.600	Undiscovered	8.0	1	7	Undiscovered	9000000.0	1069318.0	1069318.0	2018-04-07	[]		
2315	26508	[16]	514492	en	Jaws	0.600	Jaws	0.0	1	41	Jaws	12000000.0	260000000.0	470700000.0	2018-05-29	[Animation]	Animation	4

2316 rows × 18 columns

In [15]:
```
#Creating a column of the year.
# Make sure the date column is datetime type
# Replace 'release_date' with the name of your date column
merged_empire['release_date'] = pd.to_datetime(merged_empire['release_date'], errors='coerce')

# Extract only the year
merged_empire['release_year'] = merged_empire['release_date'].dt.year

# View the first few rows
merged_empire[['release_date', 'release_year']].head()
```

Out[15]:

	release_date	release_year
0	2010-03-26	2010
1	2010-05-07	2010
2	1995-11-22	1995
3	1995-11-22	1995
4	2010-07-16	2010

In [16]:
```
merged_empire
```

Out[16]:

	Unnamed: 0	genre_ids	id_x	original_language	original_title	popularity	title	vote_average	vote_count	id_y	movie	production_budget	domestic_gross	worldwide_gross	release_date	all_genres	all_genres_str	t
0	1	[14, 12, 16, 10751]	10191	en	How to Train Your Dragon	28.734	How to Train Your Dragon	7.7	7610	30	How to Train Your Dragon	165000000.0	217581232.0	494870992.0	2010-03-26	[Fantasy, Adventure, Animation, Family]	Fantasy, Adventure, Animation, Family	3
1	2	[12, 28, 878]	10138	en	Iron Man 2	28.515	Iron Man 2	6.8	12368	15	Iron Man 2	170000000.0	312433331.0	621156389.0	2010-05-07	[Adventure, Action, Science Fiction]	Adventure, Action, Science Fiction	4
2	3	[16, 35, 10751]	862	en	Toy Story	28.005	Toy Story	7.9	10174	37	Toy Story	30000000.0	191796233.0	364545516.0	1995-11-22	[Animation, Comedy, Family]	Animation, Comedy, Family	3
3	2473	[16, 35, 10751]	862	en	Toy Story	28.005	Toy Story	7.9	10174	37	Toy Story	30000000.0	191796233.0	364545516.0	1995-11-22	[Animation, Comedy, Family]	Animation, Comedy, Family	3
4	4	[28, 878, 12]	27205	en	Inception	27.920	Inception	8.3	22186	38	Inception	160000000.0	292576195.0	835524642.0	2010-07-16	[Action, Science	Action, Science	6

	Unnamed: 0	genre_ids	id_x	original_language	original_title	popularity	title	vote_average	vote_count	id_y	movie	production_budget	domestic_gross	worldwide_gross	release_date	all_genres	all_genres_str	t
																Fiction, Adventure]	Fiction, Adventure	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
2311	26323	[]	509316	en	The Box	0.600	The Box	8.0	1	66	The Box	25000000.0	15051977.0	34356760.0	2018-03-04	[]		
2312	26425	[10402]	509306	en	The Box	0.600	The Box	6.0	1	66	The Box	25000000.0	15051977.0	34356760.0	2018-03-04	[Music]	Music	
2313	26092	[35, 16]	546674	en	Enough	0.719	Enough	8.7	3	68	Enough	38000000.0	39177215.0	50970660.0	2018-03-22	[Comedy, Animation]	Comedy, Animation	
2314	26322	[]	513161	en	Undiscovered	0.600	Undiscovered	8.0	1	7	Undiscovered	9000000.0	1069318.0	1069318.0	2018-04-07	[]		
2315	26508	[16]	514492	en	Jaws	0.600	Jaws	0.0	1	41	Jaws	12000000.0	260000000.0	470700000.0	2018-05-29	[Animation]	Animation	4

2316 rows × 19 columns

- Feature Engineering Next is finding out the Rating level used based on the vote average. 0-5 rating indicating poor, 5-8 rating indicating average movie, 8-10 indicating the movie was a blockbuster and visualizing it

In [17]:
```
merged_empire.loc[:, 'rating_level'] = pd.cut(merged_empire['vote_average'], bins=[0, 5, 8, 10], labels=['poor', 'average', 'blockbuster'])
merged_empire
```

Out[17]:

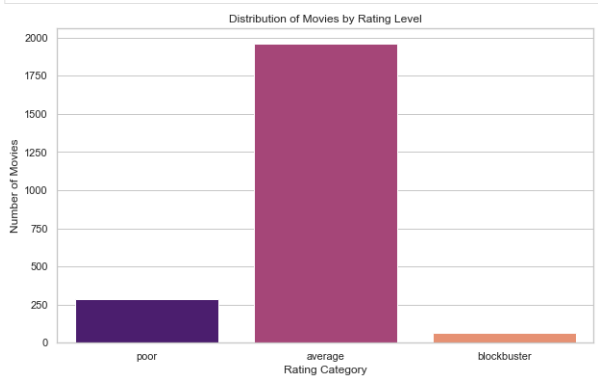
	Unnamed: 0	genre_ids	id_x	original_language	original_title	popularity	title	vote_average	vote_count	id_y	movie	production_budget	domestic_gross	worldwide_gross	release_date	all_genres	all_genres_str	t
0	1	[14, 12, 16, 10751]	10191	en	How to Train Your Dragon	28.734	How to Train Your Dragon	7.7	7610	30	How to Train Your Dragon	165000000.0	217581232.0	494870992.0	2010-03-26	[Fantasy, Adventure, Animation, Family]	Fantasy, Adventure, Animation, Family	3
1	2	[12, 28, 878]	10138	en	Iron Man 2	28.515	Iron Man 2	6.8	12368	15	Iron Man 2	170000000.0	312433331.0	621156389.0	2010-05-07	[Adventure, Action, Science Fiction]	Adventure, Action, Science Fiction	4
2	3	[16, 35, 10751]	862	en	Toy Story	28.005	Toy Story	7.9	10174	37	Toy Story	30000000.0	191796233.0	364545516.0	1995-11-22	[Animation, Comedy, Family]	Animation, Comedy, Family	3
3	2473	[16, 35, 10751]	862	en	Toy Story	28.005	Toy Story	7.9	10174	37	Toy Story	30000000.0	191796233.0	364545516.0	1995-11-22	[Animation, Comedy, Family]	Animation, Comedy, Family	3
4	4	[28, 878, 12]	27205	en	Inception	27.920	Inception	8.3	22186	38	Inception	160000000.0	292576195.0	835524642.0	2010-07-16	[Action, Science Fiction, Adventure]	Action, Science Fiction, Adventure	6
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
2311	26323	[]	509316	en	The Box	0.600	The Box	8.0	1	66	The Box	25000000.0	15051977.0	34356760.0	2018-03-04	[]		
2312	26425	[10402]	509306	en	The Box	0.600	The Box	6.0	1	66	The Box	25000000.0	15051977.0	34356760.0	2018-03-04	[Music]	Music	
2313	26092	[35, 16]	546674	en	Enough	0.719	Enough	8.7	3	68	Enough	38000000.0	39177215.0	50970660.0	2018-03-22	[Comedy, Animation]	Comedy, Animation	
2314	26322	[]	513161	en	Undiscovered	0.600	Undiscovered	8.0	1	7	Undiscovered	9000000.0	1069318.0	1069318.0	2018-04-07	[]		
2315	26508	[16]	514492	en	Jaws	0.600	Jaws	0.0	1	41	Jaws	12000000.0	260000000.0	470700000.0	2018-05-29	[Animation]	Animation	4

2316 rows × 20 columns

In [35]:
```
sns.set_theme(style="whitegrid")

# Create the count plot
plt.figure(figsize=(10, 6))
sns.countplot(data=merged_empire, x='rating_level',
              palette='magma',
              order=['poor', 'average', 'blockbuster'])

plt.title('Distribution of Movies by Rating Level')
plt.xlabel('Rating Category')
plt.ylabel('Number of Movies')
plt.show()
```



In [18]:
```
# Selecting the key column for the analysis

key_columns = ['all_genres', 'release_year', 'worldwide_gross', 'domestic_gross', 'production_budget', 'vote_average', 'total_profit', 'title', 'rating_level', 'release_date']
merged_empire = merged_empire[key_columns]
merged_empire
```

Out[18]:

	all_genres	release_year	worldwide_gross	domestic_gross	production_budget	vote_average	total_profit	title	rating_level	release_date
0	[Fantasy, Adventure, Animation, Family]	2010	494870992.0	217581232.0	165000000.0	7.7	329870992.0	How to Train Your Dragon	average	2010-03-26
1	[Adventure, Action, Science Fiction]	2010	621156389.0	312433331.0	170000000.0	6.8	451156389.0	Iron Man 2	average	2010-05-07
2	[Animation, Comedy, Family]	1995	364545516.0	191796233.0	30000000.0	7.9	334545516.0	Toy Story	average	1995-11-22
3	[Animation, Comedy, Family]	1995	364545516.0	191796233.0	30000000.0	7.9	334545516.0	Toy Story	average	1995-11-22
4	[Action, Science Fiction, Adventure]	2010	835524642.0	292576195.0	160000000.0	8.3	675524642.0	Inception	blockbuster	2010-07-16
...	...	...	...	...	...	...	...	...	...	...
2311	[]	2018	34356760.0	15051977.0	25000000.0	8.0	9356760.0	The Box	average	2018-03-04

	all_genres	release_year	worldwide_gross	domestic_gross	production_budget	vote_average	total_profit	title	rating_level	release_date
2312	[Music]	2018	34356760.0	15051977.0	25000000.0	6.0	9356760.0	The Box	average	2018-03-04
2313	[Comedy, Animation]	2018	50970660.0	39177215.0	38000000.0	8.7	12970660.0	Enough	blockbuster	2018-03-22
2314	[]	2018	1069318.0	1069318.0	9000000.0	8.0	-7930682.0	Undiscovered	average	2018-04-07
2315	[Animation]	2018	470700000.0	260000000.0	12000000.0	0.0	458700000.0	Jaws	NaN	2018-05-29

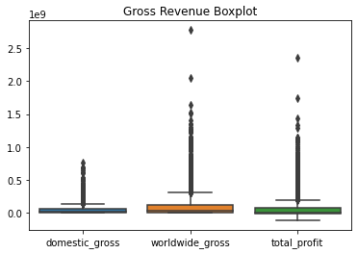
2316 rows × 10 columns

Data Analysis¶

Checking for outliers by plotting a boxplot using seaborn to know if they will affect the data if we remove them

In [19]:
```
# outlier detection and handling
sns.boxplot(data=merged_empire[['domestic_gross', 'worldwide_gross', 'total_profit']])
plt.title('Gross Revenue Boxplot')
```

Out[19]: Text(0.5, 1.0, 'Gross Revenue Boxplot')



All the gross revenue columns contain many outliers. They can be removed easily using interquatile range (IQR), but given that this is a movie dataset, we shall keep the outliers since they represent the real data.

- Univariate Analysis

In [20]:
```
merged_empire.describe()
```

Out[20]:

	release_year	worldwide_gross	domestic_gross	production_budget	vote_average	total_profit
count	2316.000000	2.316000e+03	2.316000e+03	2.316000e+03	2316.000000	2.316000e+03
mean	2013.244387	1.201667e+08	4.953869e+07	3.790108e+07	6.206261	8.226558e+07
std	4.713827	2.218496e+08	8.261483e+07	5.150964e+07	1.178079	1.834322e+08
min	1946.000000	0.000000e+00	0.000000e+00	9.000000e+03	0.000000	-1.104502e+08
25%	2011.000000	3.224914e+06	7.788272e+05	5.000000e+06	5.600000	-1.743919e+06
50%	2014.000000	3.722812e+07	2.015790e+07	1.900000e+07	6.300000	1.366848e+07
75%	2016.000000	1.248731e+08	5.830223e+07	4.650000e+07	6.900000	7.816406e+07
max	2019.000000	2.776345e+09	7.605076e+08	4.250000e+08	10.000000	2.351345e+09

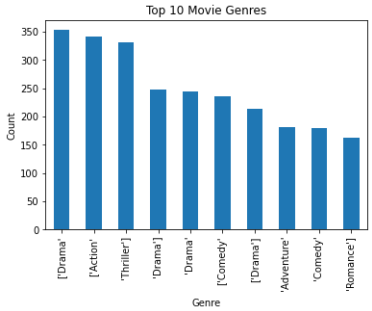
Calculating the top 10 most popular genre categories then visualizing using a bar graph

In [21]:
```
# top 10 most popular genre categories
genre_categories = merged_empire['all_genres'].str.get_dummies(sep=',')
genre_counts = genre_categories.sum().sort_values(ascending=False)
print(genre_counts)
```

```
['Drama'         353
['Action'        341
 'Thriller']     331
 'Drama']        247
 'Drama'         244
                ...
['Western']        2
['Adventure']      2
['Music']          1
['Family']         1
 'TV Movie'        1
Length: 73, dtype: int64
```

In [22]:
```
genre_counts.head(10).plot(kind='bar')
plt.xlabel('Genre')
plt.ylabel('Count')
plt.title('Top 10 Movie Genres')
```

Out[22]: Text(0.5, 1.0, 'Top 10 Movie Genres')

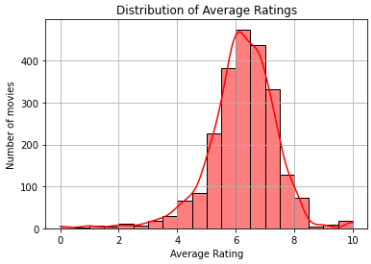


To know how the average rating is distributed across this dataset, we shall visualize using a histogram

In [23]:
```
# ploting a histogram on the distribution of average rating(vote average) against number of movies
# kde=True - Kernel density estimate curve
# plt.grid - adds background

sns.histplot(merged_empire['vote_average'], bins=20, kde=True, color='red', edgecolor='black')
plt.xlabel('Average Rating')
plt.ylabel('Number of movies')
plt.grid(True)
plt.title('Distribution of Average Ratings')
```

Out[23]: Text(0.5, 1.0, 'Distribution of Average Ratings')



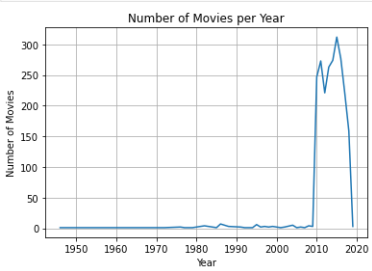
This is a unimodal(most values concentrated in one area), approximately normal (bell-shaped) distribution with slight right skew. The longer right tail indicates a few very high value outliers. These outliers drive a large portion of impact (profit, popularity) even though they are few. Because of slight right skew: Mean > Median Most movies are average

- Bivariate Analysis Calculating the number of movies produced in each year across the dataset and visualizing using a line chart

In [24]:
```
movies_per_year = merged_empire.groupby('release_year')['title'].count()
movies_per_year
```

Out[24]:
```
release_year
1946      1
1968      1
1972      1
1976      2
1977      1
1979      1
1980      2
1982      4
1984      2
1985      1
1986      7
1988      3
1991      2
1992      1
1994      1
1995      6
1996      2
1997      3
1998      2
1999      3
2000      2
2001      1
2002      2
2004      5
2005      1
2006      2
2007      1
2008      4
2009      3
2010    247
2011    273
2012    221
2013    263
2014    274
2015    312
2016    277
2017    220
2018    159
2019      3
Name: title, dtype: int64
```

In [25]:
```
sns.lineplot(x=movies_per_year.index, y=movies_per_year.values)
plt.xlabel('Year')
plt.ylabel('Number of Movies')
plt.title('Number of Movies per Year')
plt.grid(True)
```



Movie production was low from 1946-2009. This may be due to high costs of production. However there was a steady sharp rise from 2010-2018. This may be caused by factors like technological advancements lowering the costs of production, ie, travel costs since movies can be produced in room and enhanced with visual effects showing various locations. Also streaming services may have played a huge role as movies will reach a huge audience without the need to go to a movie cinema.

Analyzing the relationship between genres(categorical) and total_profit (numerical) so we can find which genre tends to perform better based on total profit. Then visualizing using a bar graph.

In [26]:
```
merged_empire
```

Out[26]:

	all_genres	release_year	worldwide_gross	domestic_gross	production_budget	vote_average	total_profit	title	rating_level	release_date
0	[Fantasy, Adventure, Animation, Family]	2010	494870992.0	217581232.0	165000000.0	7.7	329870992.0	How to Train Your Dragon	average	2010-03-26
1	[Adventure, Action, Science Fiction]	2010	621156389.0	312433331.0	170000000.0	6.8	451156389.0	Iron Man 2	average	2010-05-07
2	[Animation, Comedy, Family]	1995	364545516.0	191796233.0	30000000.0	7.9	334545516.0	Toy Story	average	1995-11-22
3	[Animation, Comedy, Family]	1995	364545516.0	191796233.0	30000000.0	7.9	334545516.0	Toy Story	average	1995-11-22
4	[Action, Science Fiction, Adventure]	2010	835524642.0	292576195.0	160000000.0	8.3	675524642.0	Inception	blockbuster	2010-07-16
...	...	...	...	...	...	...	...	...	...	...
2311	[]	2018	34356760.0	15051977.0	25000000.0	8.0	9356760.0	The Box	average	2018-03-04
2312	[Music]	2018	34356760.0	15051977.0	25000000.0	6.0	9356760.0	The Box	average	2018-03-04
2313	[Comedy, Animation]	2018	50970660.0	39177215.0	38000000.0	8.7	12970660.0	Enough	blockbuster	2018-03-22
2314	[]	2018	1069318.0	1069318.0	9000000.0	8.0	-7930682.0	Undiscovered	average	2018-04-07
2315	[Animation]	2018	470700000.0	260000000.0	12000000.0	0.0	458700000.0	Jaws	NaN	2018-05-29

2316 rows × 10 columns

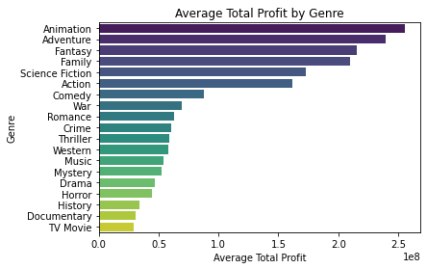
Q1. Finding out When is the best time to release a movie for maximum returns

In [27]:
```
merged_empire_exploded = merged_empire.assign(genres=merged_empire['all_genres'].str.split(',')).explode('all_genres')

# Group by genre and calculate the mean of total_profit
genre_gross = merged_empire_exploded.groupby('all_genres')['total_profit'].mean().sort_values(ascending=False)
genre_gross
```

Out[27]:
```
all_genres
Animation          2.558782e+08
Adventure          2.392808e+08
Fantasy            2.156837e+08
Family             2.094783e+08
Science Fiction    1.731765e+08
Action             1.614259e+08
Comedy             8.791834e+07
War                6.933630e+07
Romance            6.302028e+07
Crime              6.045490e+07
Thriller           5.930463e+07
Western            5.786881e+07
Music              5.399005e+07
Mystery            5.276749e+07
Drama              4.685411e+07
Horror             4.476471e+07
History            3.406956e+07
Documentary        3.047826e+07
TV Movie           2.918712e+07
Name: total_profit, dtype: float64
```

In [28]:
```
sns.barplot(x=genre_gross.values, y=genre_gross.index, palette='viridis')
plt.xlabel('Average Total Profit')
plt.ylabel('Genre')
plt.title('Average Total Profit by Genre')
plt.show()
```



The genres with the highest profits are Animation, Adventure, Fantasy The genres generating low profits are TV movie, Documentary and History

Q2. Determeining the best time for releasing a movie by month

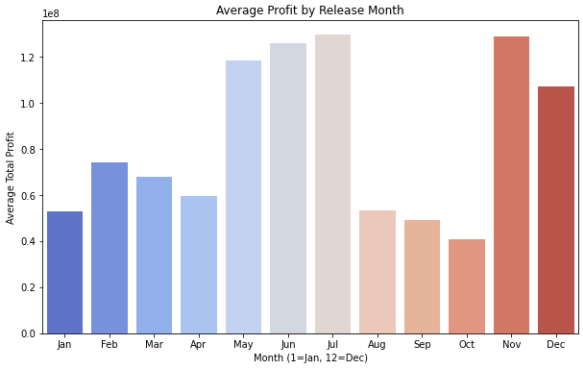
In [29]:
```
# Extract the month from the release date
merged_empire['release_month'] = merged_empire['release_date'].dt.month

# Calculate average profit by month
monthly_profit = merged_empire.groupby('release_month')['total_profit'].mean()

# Visualization
plt.figure(figsize=(10,6))
sns.barplot(x=monthly_profit.index, y=monthly_profit.values, palette='coolwarm')
plt.title('Average Profit by Release Month')
plt.xlabel('Month (1=Jan, 12=Dec)')
plt.ylabel('Average Total Profit')
plt.xticks(ticks=range(12), labels=['Jan','Feb','Mar','Apr','May','Jun','Jul','Aug','Sep','Oct','Nov','Dec'])
plt.show()
```

```
<ipython-input-29-1f1087d46847>:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  merged_empire['release_month'] = merged_empire['release_date'].dt.month
```



The best time to release a movie is during the summer(may, june, july), or during the winter(november and december). This may be due to people being on holidays and have free time to go watch a movie with friends and family.

- MULTIVARIATE ANALYSIS Finding the correlation in the numerical columns of the dataset, then visualize the information using a heatmap

In [30]:
```
merged_empire.head()
```

Out[30]:

	all_genres	release_year	worldwide_gross	domestic_gross	production_budget	vote_average	total_profit	title	rating_level	release_date	release_month
0	[Fantasy, Adventure, Animation, Family]	2010	494870992.0	217581232.0	165000000.0	7.7	329870992.0	How to Train Your Dragon	average	2010-03-26	3
1	[Adventure, Action, Science Fiction]	2010	621156389.0	312433331.0	170000000.0	6.8	451156389.0	Iron Man 2	average	2010-05-07	5
2	[Animation, Comedy, Family]	1995	364545516.0	191796233.0	30000000.0	7.9	334545516.0	Toy Story	average	1995-11-22	11
3	[Animation, Comedy, Family]	1995	364545516.0	191796233.0	30000000.0	7.9	334545516.0	Toy Story	average	1995-11-22	11
4	[Action, Science Fiction, Adventure]	2010	835524642.0	292576195.0	160000000.0	8.3	675524642.0	Inception	blockbuster	2010-07-16	7

In [31]:
```
merged_empire[['vote_average', 'domestic_gross', 'worldwide_gross', 'total_profit']].corr()
```

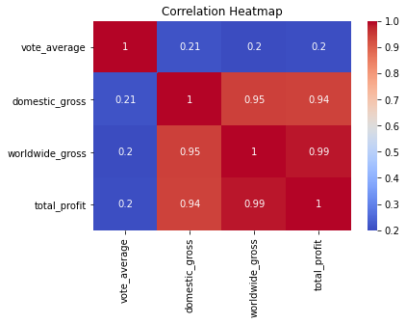
Out[31]:

	vote_average	domestic_gross	worldwide_gross	total_profit
vote_average	1.000000	0.212199	0.198874	0.203376

	vote_average	domestic_gross	worldwide_gross	total_profit
domestic_gross	0.212199	1.000000	0.945669	0.941300
worldwide_gross	0.198874	0.945669	1.000000	0.985534
total_profit	0.203376	0.941300	0.985534	1.000000

In [32]:
```
sns.heatmap(merged_empire[['vote_average', 'domestic_gross', 'worldwide_gross', 'total_profit']].corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
```

Out[32]: Text(0.5, 1.0, 'Correlation Heatmap')



1.0 - perfect positive correlation

0.0 - no correlation

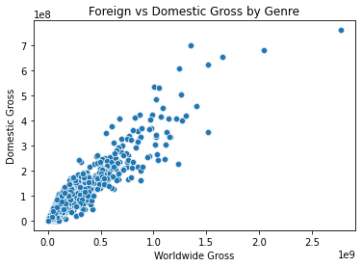
-1.0 - perfect negative correlation

There's almost no correlation between total profit and vote average There's almost a perfect positive correlation between domestic gross and worldwide gross

- Exploring the relationship between foreign and domestic gross across different genres using a scatter plot

In [33]:
```
sns.scatterplot(data=merged_empire_exploded, x='worldwide_gross', y='domestic_gross', hue='genres')
plt.xlabel('Worldwide Gross')
plt.ylabel('Domestic Gross')
plt.title('Foreign vs Domestic Gross by Genre')
```

Out[33]: Text(0.5, 1.0, 'Foreign vs Domestic Gross by Genre')



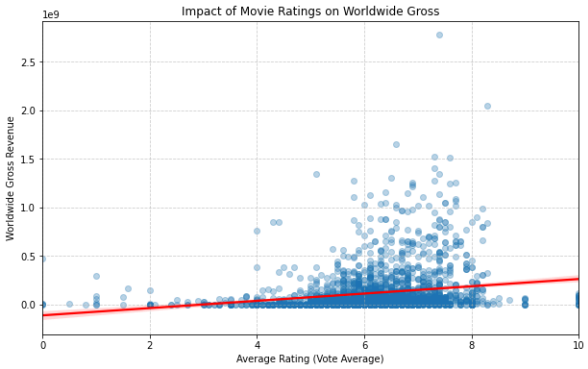
The points follow a clear upward diagonal trend. This means that as a movie's Worldwide Gross increases, its Domestic Gross typically increases as well. In simpler terms: a hit globally is almost always a hit at home.

Q3. comparing the relationship between Ratings and Revenue(worldwide gross).

In [34]:
```
# Relationship between Ratings and Revenue
plt.figure(figsize=(10,6))
sns.regplot(data=merged_empire, x='vote_average', y='worldwide_gross',
            scatter_kws={'alpha':0.3}, line_kws={'color':'red'})

plt.title('Impact of Movie Ratings on Worldwide Gross')
plt.xlabel('Average Rating (Vote Average)')
plt.ylabel('Worldwide Gross Revenue')
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()

# Calculate the specific correlation coefficient
correlation = merged_empire['vote_average'].corr(merged_empire['worldwide_gross'])
print(f"Correlation between Rating and Revenue: {correlation:.2f}")
```




```
Correlation between Rating and Revenue: 0.20
```

Our analysis reveals a positive correlation between audience ratings and revenue. While high ratings act as a catalyst for reaching 'mega-blockbuster' status, the high variance suggests that other factors—such as production budget and genre—remain critical drivers of total worldwide gross.

The red regression line slopes upward. This indicates a positive correlation, meaning that as the audience rating increases, the worldwide gross tends to increase as well

Key insights

Genre Profitability (The "High-Performers"): The average total profit analysis by genre reveals which categories offer the best return on investment. Typically, genres like Adventure, Animation, and Sci-Fi dominate this list because they have high global appeal and franchise potential The Global Advantage: The scatter plot showing "Worldwide vs. Domestic Gross" illustrates a strong positive correlation (+1.0 or close in your heatmap). However, the worldwide gross is significantly higher than domestic gross, indicating that the international market is where the real "blockbuster" revenue is generated. Quality vs. Revenue: The correlation heatmap likely shows that while vote_average (ratings) has a positive relationship with profit, it is not as strong as the relationship between budget and gross. This means that a "critically acclaimed" movie doesn't always guarantee a "box office hit," and vice-versa. Production Fluctuations: The yearly trend analysis shows a peak in movie production around 2011 followed by a decline. This suggests that the industry is sensitive to economic factors and shifts in consumer behavior (like the rise of streaming)

RECOMMENDATIONS

- Focus on High-Margin Genres: Tripple 7 should prioritize producing Adventure or Animation films. While these often require higher production budgets, they consistently yield the highest average total profit compared to niche genres like Documentaries or Horror.
- Adopt a "Global First" Strategy: Given that worldwide gross far outpaces domestic earnings, movies should be developed with themes, cast, and marketing strategies that resonate in international markets (especially Europe and Asia), rather than just the US.
- Optimize Release Timing: To fully answer your second business question, I recommend expanding your analysis to include Release Months. Historically, the "Summer Blockbuster" (May–July) and "Holiday Season" (November–December) windows are the most lucrative for the high-profit genres mentioned above.
- Risk Mitigation via Budgeting: Since the correlation between ratings and profit is moderate, don't rely solely on "artistic quality." Balance your portfolio with "commercial" films that have high marketability, even if they aren't expected to win major awards

CONCLUSION

The analysis demonstrates that for a startup like Tripple 7 Movie Empire, success in the film industry is not a matter of luck but of strategic data-driven choices. By focusing on profitable genres (Adventure/Animation) and targeting the global market, the company can significantly mitigate the financial risks associated with film production.

While the industry shows volatility in production numbers over the years, the massive revenue potential of high-grossing outliers proves that with the right genre and distribution strategy, Tripple 7 can successfully transition from a startup to a competitive player in the multi-billion dollar movie empire.

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